

VLSI Design & Practice 04

22-bit 2-stage pipelined
RCA, CSA and SRCSA

Spring semester, 2026

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Contents

- 1. 2-stage pipelined 22-bit Ripple Carry Adder**
 - Do the pipelining with the Gate level description design & Check the functionality
 - Measure the critical path delay & circuit area & power
- 2. 2-stage pipelined 20-bit Carry Select Adder**
 - Do the pipelining with the Gate level description & Check the functionality
 - Measure the critical path delay & circuit area & power
- 3. 2-stage pipelined 22-bit Square Root Carry Select Adder**
 - Do the pipelining with the Gate level description & Check the functionality
 - Measure the critical path delay & circuit area & power
- 4. Synthesis Results Analysis**
 - Check the experimental results (max. clock cycle, area, power)

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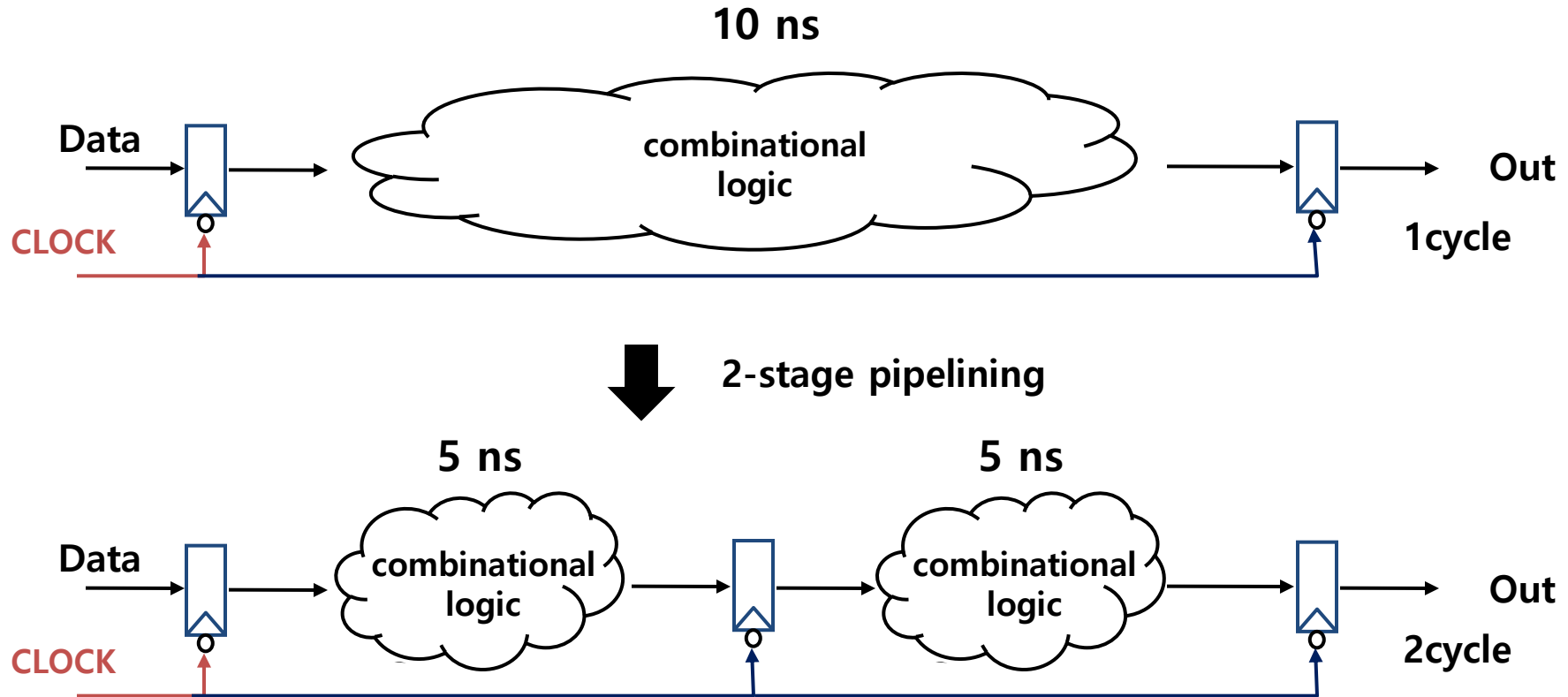
3. 2-stage pipelined 22-bit Square Root Carry Select Adder

- Do the pipelining with the Gate level description & Check the functionality
- Measure the critical path delay & circuit area & power

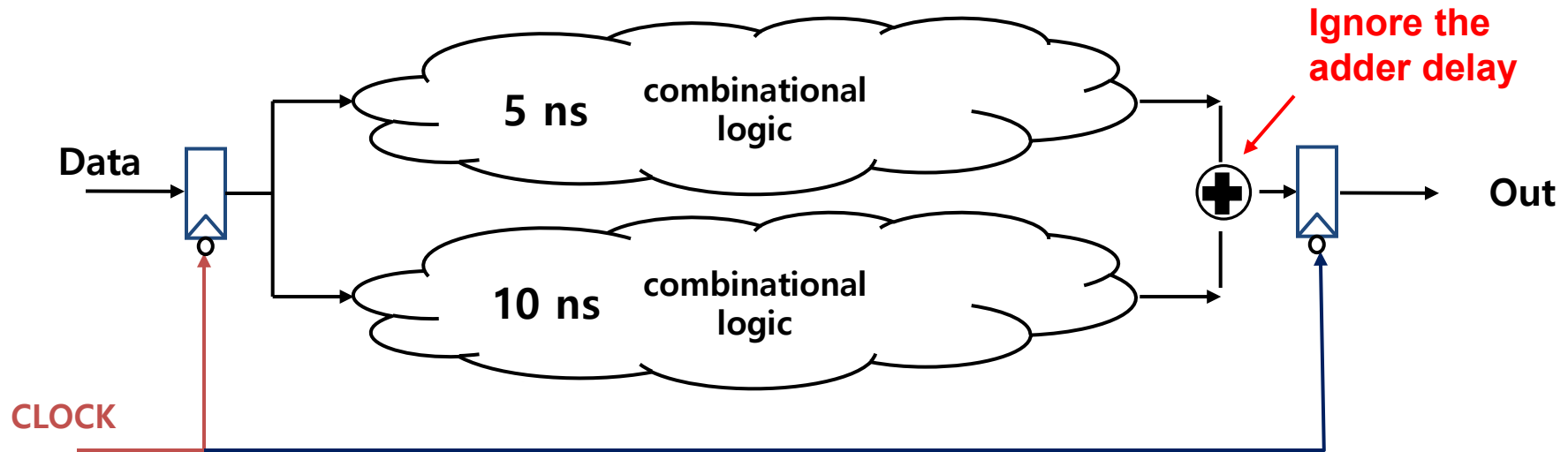
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- Check the experimental results (max. clock cycle, area, power)

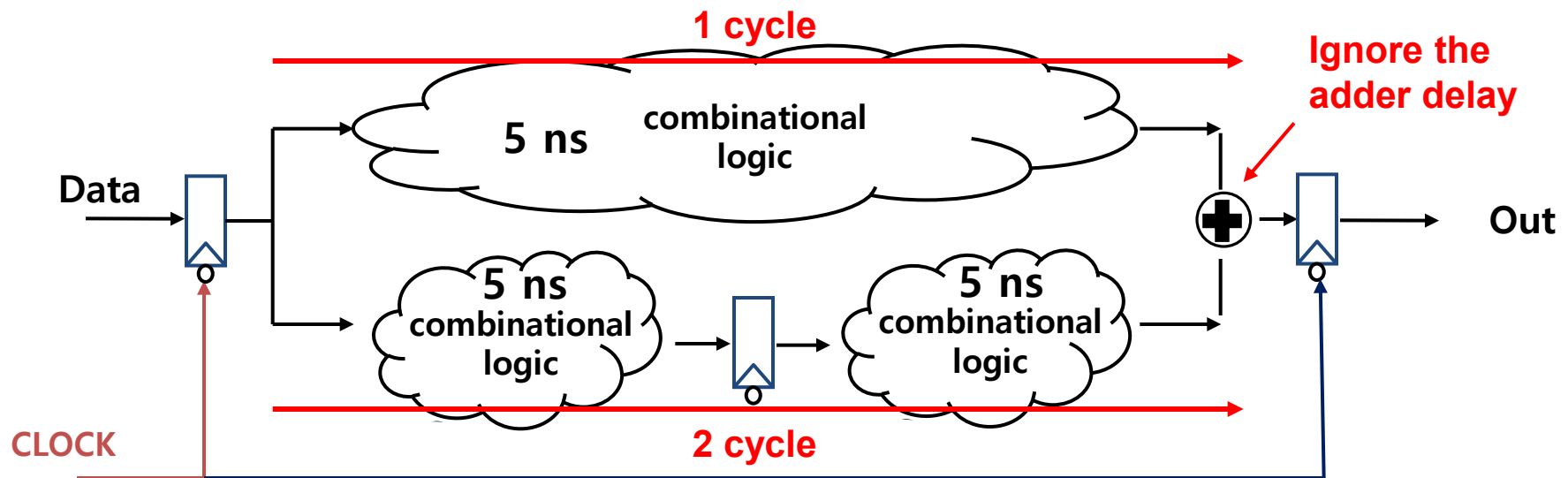
2-Stage Pipelining



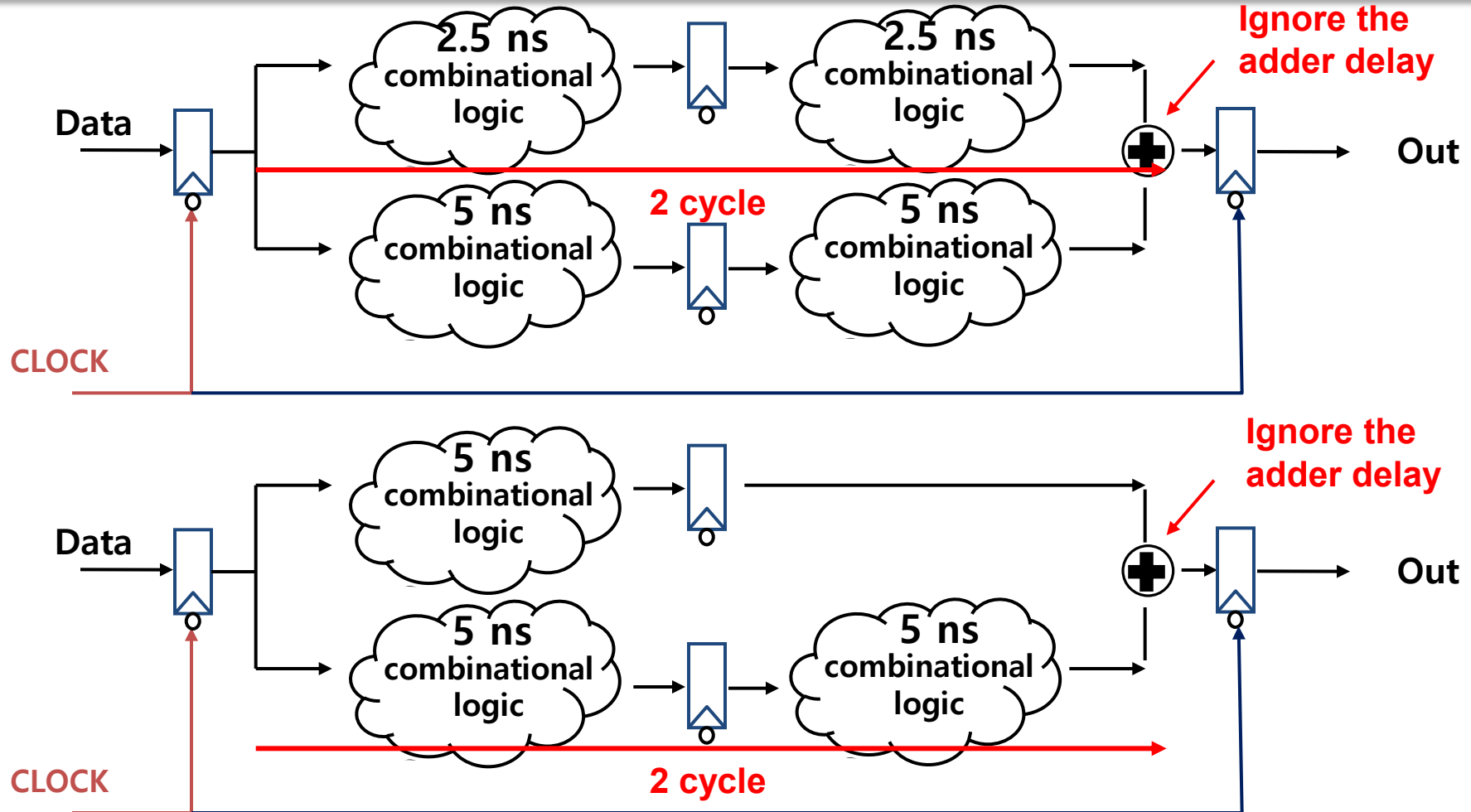
2-Stage Pipelining



2-Stage Pipelining

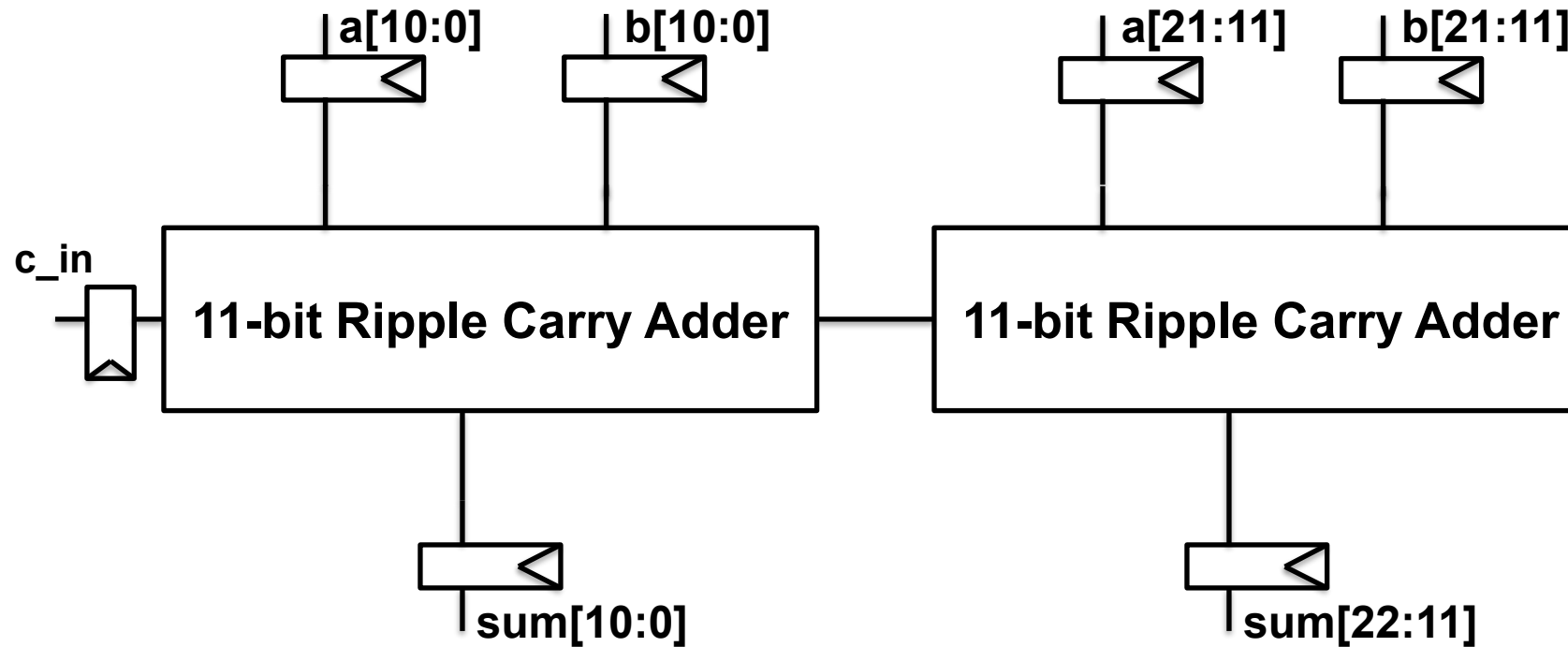


2-Stage Pipelining



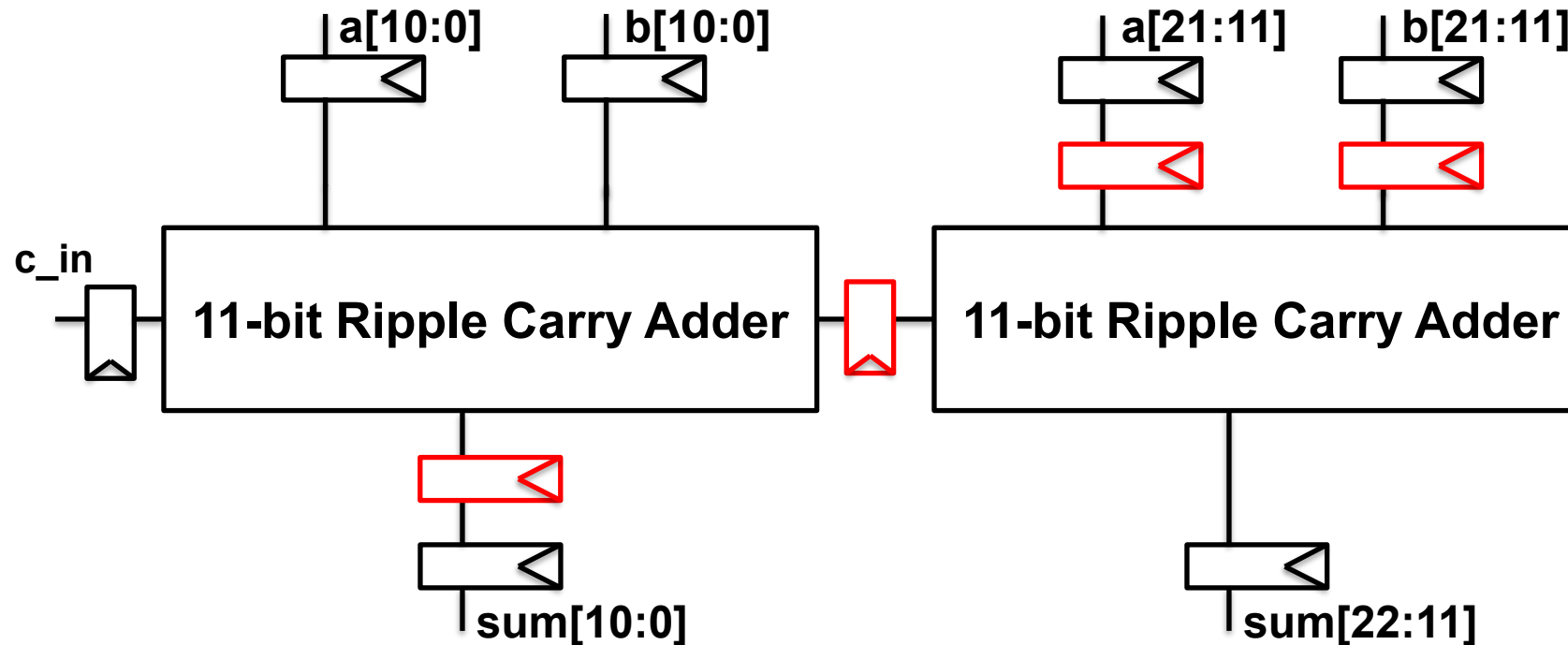
2-stage pipelined Ripple Carry Adder

Design 2-stage pipelined 22-bit Ripple Carry Adder



2-stage pipelined Ripple Carry Adder

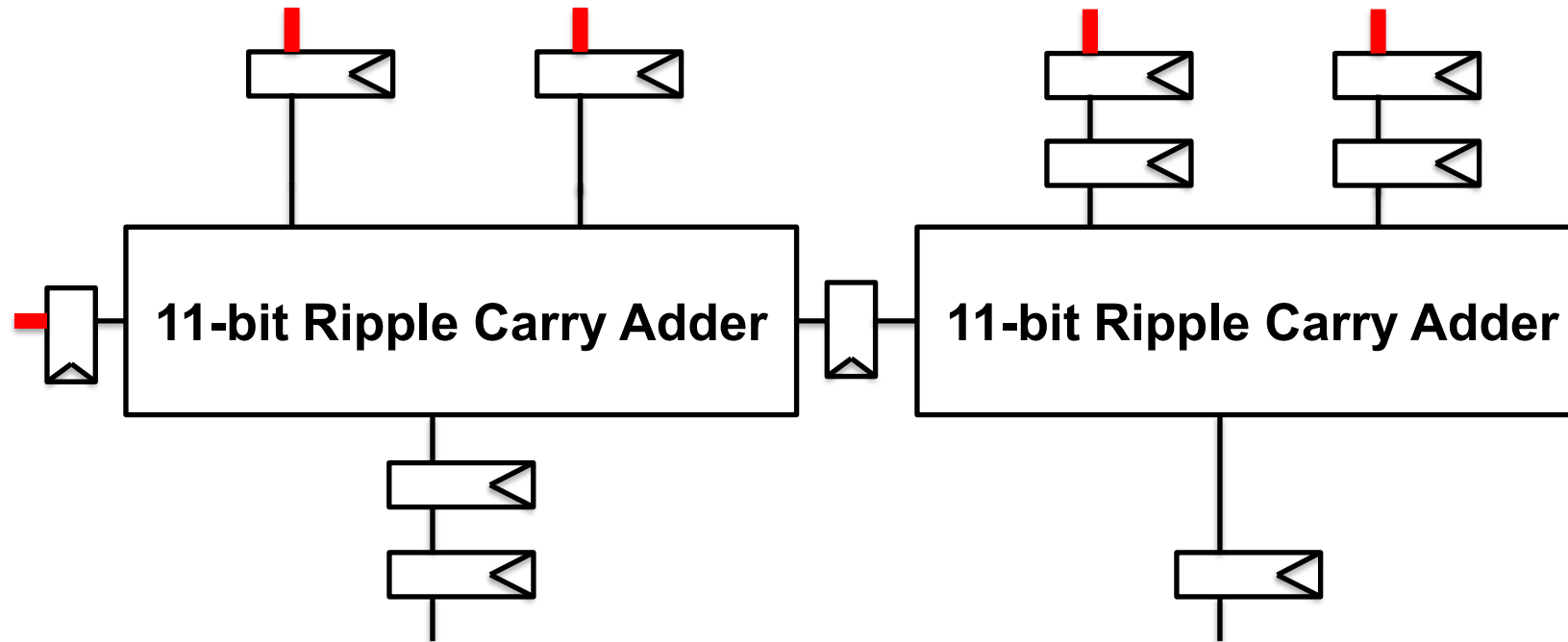
Design 2-stage pipelined 22-bit Ripple Carry Adder



1. Add D flip-flop in middle point
2. Add D flip-flop in output port of LSB side
3. Add D flip-flop in input port of MSB side

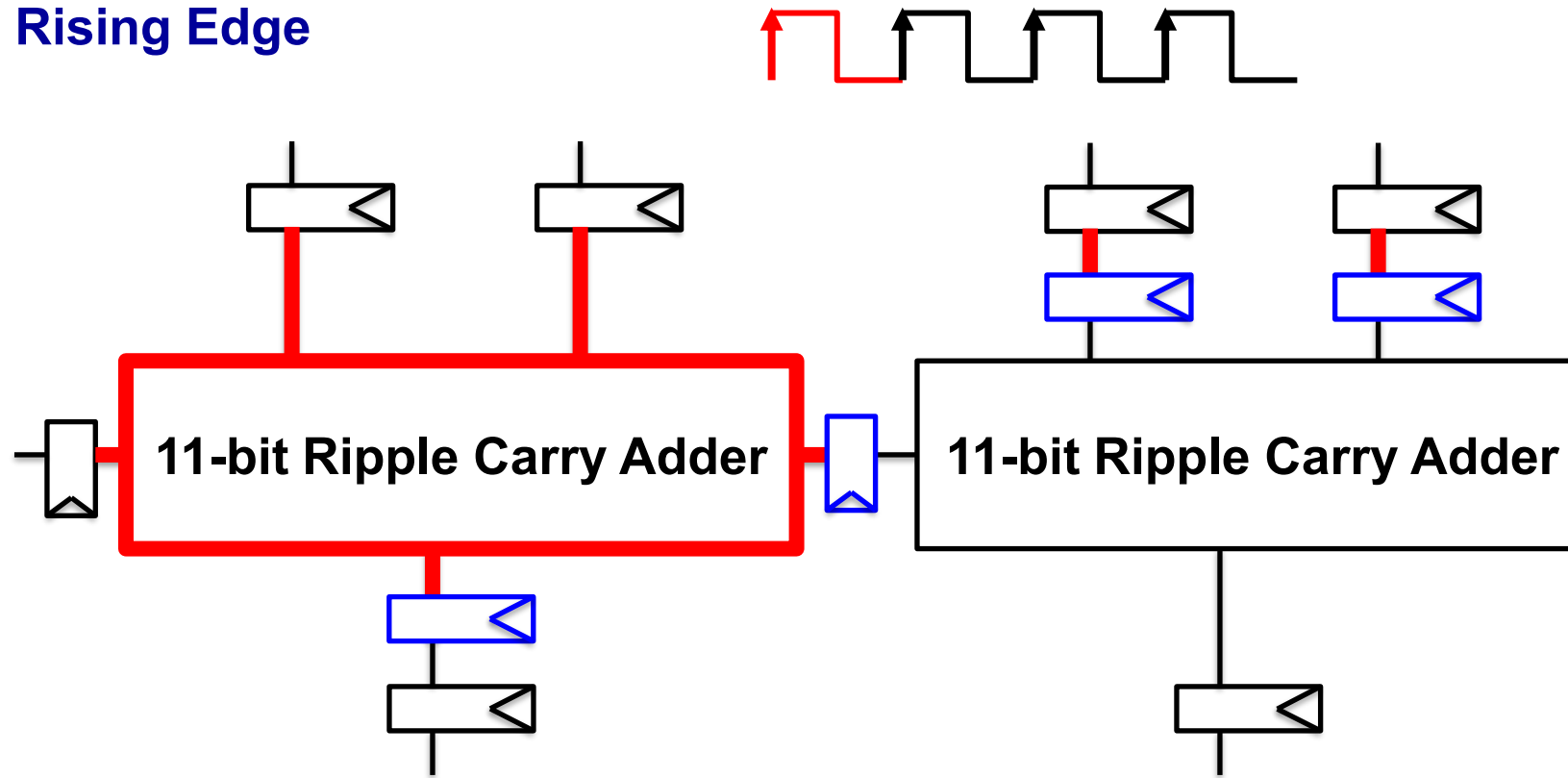
2-stage pipelined Ripple Carry Adder

Initial Condition



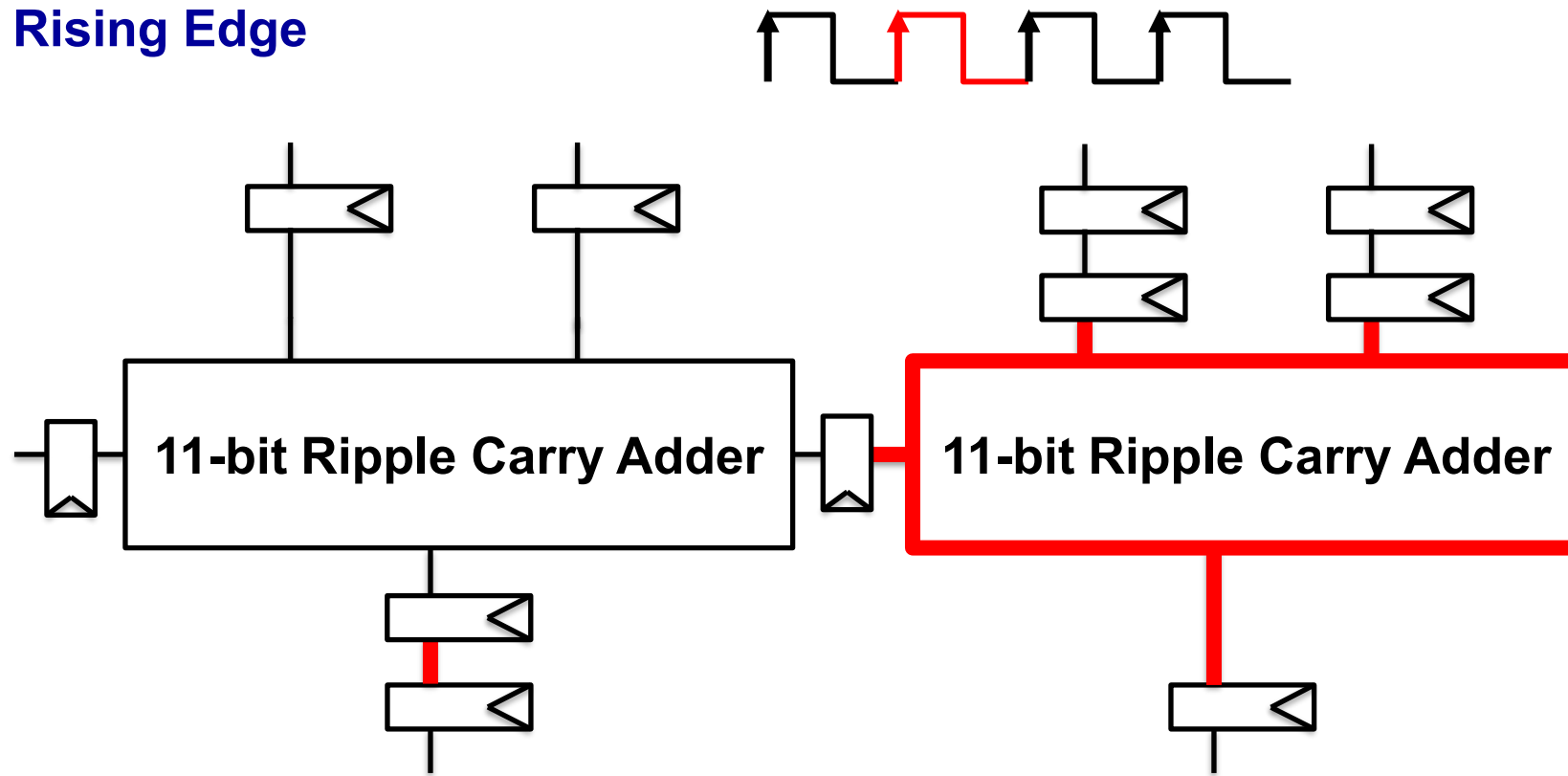
2-stage pipelined Ripple Carry Adder

0th Rising Edge



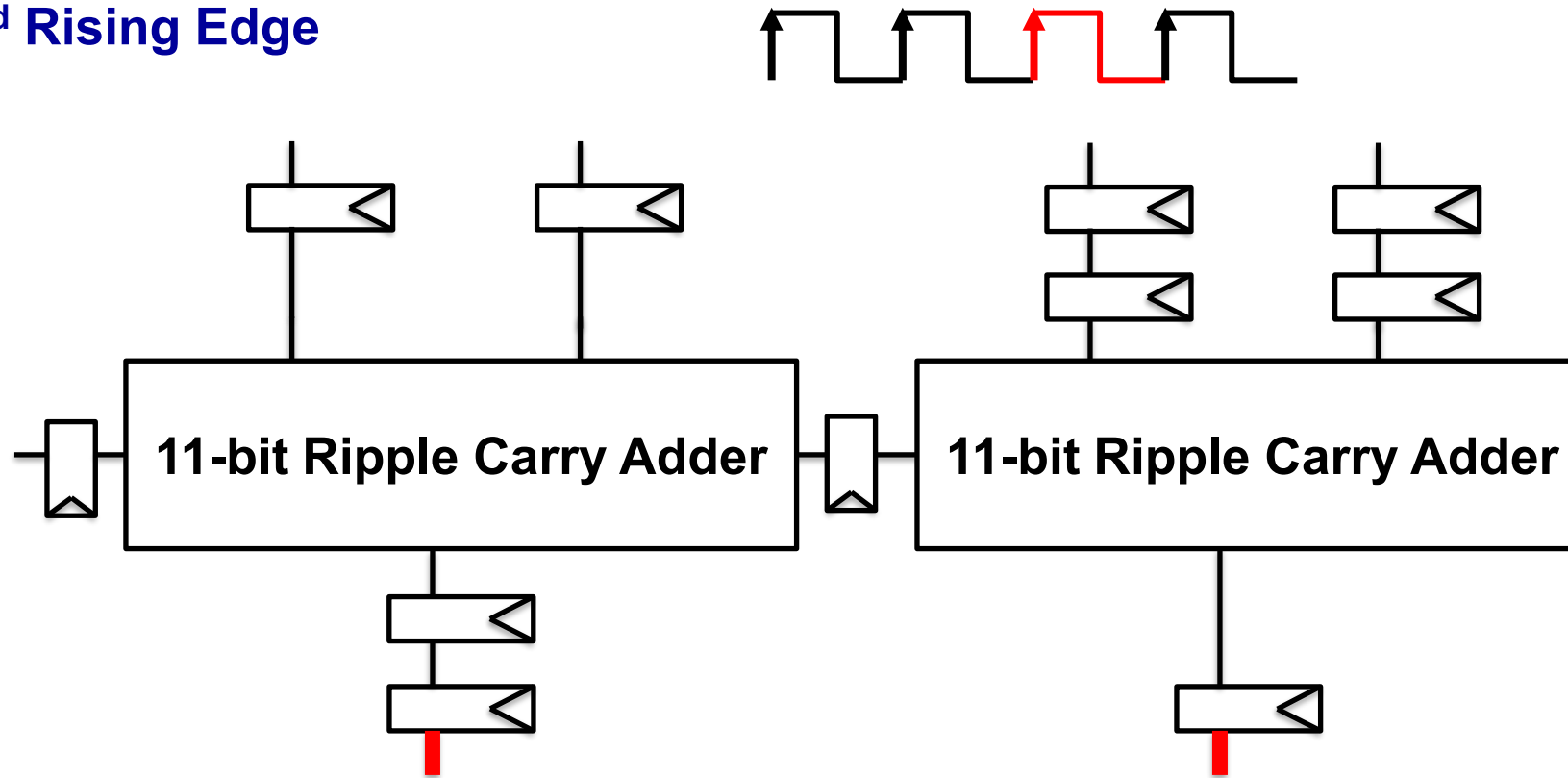
2-stage pipelined Ripple Carry Adder

1st Rising Edge



2-stage pipelined Ripple Carry Adder

2nd Rising Edge



Functional Verifications

Integrated version of stimulus file

```
1  `timescale 1 ns / 1 ps
2  module stimulus_sqrt_csa_28bit_2stage;
3
4      wire    [28:0]  sum_pipe;
5      reg     [27:0]  a, b;
6      reg     clk;
7      reg     rstn;
8
9      reg     [27:0]  mat_a [0:99];
10     reg     [27:0]  mat_b [0:99];
11     reg     [28:0]  mat_sum [0:99];
12     reg     [28:0]  mat_sum_cmp;
13
14     integer i;
15     integer k;
16     integer err;
17
18     sqrt_csa_28bit_2stage ADD0(.sum(sum_pipe), .a(
19
20     always #5 clk <= ~clk;
21
22     initial begin
23         clk <= 1; rstn <= 0;
24         #12 rstn <= 1;
25         #1050 $stop;
26     end
```

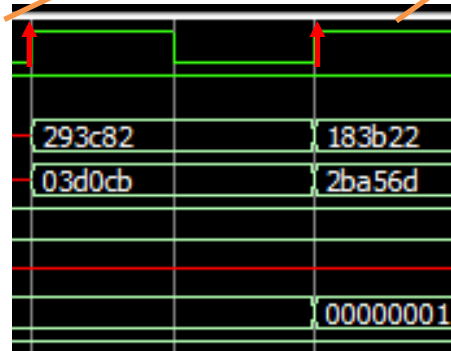
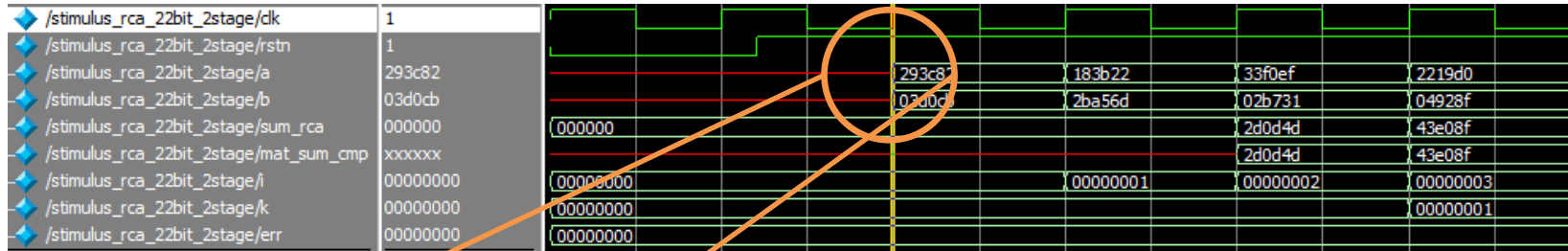
**You can modify
stimulus if needed!**

```
28  initial begin
29      $readmemh("a_input_28b.txt", mat_a);
30      $readmemh("b_input_28b.txt", mat_b);
31
32      i = 0;
33      #(20);
34      for(i = 0; i < 100; i = i + 1) begin
35          a = mat_a[i];
36          b = mat_b[i];
37          #(10);
38      end
39  end
40
41  initial begin
42      $readmemh("sum_output_29b.txt", mat_sum);
43
44      k = 0;
45      err = 0;
46      #(40);
47      for(k = 0; k < 100; k = k + 1) begin
48          mat_sum_cmp = mat_sum[k];
49          #(2);
50          if(sum_pipe != mat_sum_cmp) begin
51              err = err + 1;
52          end
53          #(8);
54      end
55      $stop;
56  end
57
58  endmodule
```

- Check outputs of adder and Matlab results are 'same'
- Check the value of error counting number equals 'zero'

Clock (posedge)

Simulation file is adjusted to positive edge

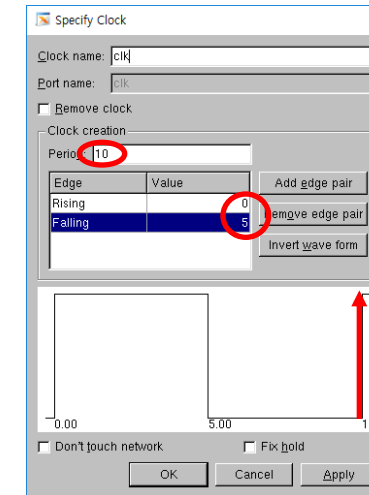


Use posedge clk!

always@(**posedge** clk)

...

In Modelsim



In Design Compiler

Tip) Report Clock Period

① Select 'CLK' in heirarchy window.

② Attributes / Specify Clock

③ Clock name

④ Period

⑤ Rising point & Falling point

create_clock -name "clk" -period 2 -waveform { 1 2 } { clk }

Specify Clock

Clock name: clk

Port name: clk

☐ Remove clock

Clock creation

Period: 2

Edge	Value
Rising	1.000
Falling	2.000

☐ Don't touch network ☐ Fix hold

Cancel Apply

Tip) Report Clock Period

```
create_clock -name "clk" -period 5.7 -waveform { 2.850 5.700 } { clk }
```

Startpoint: b_q_reg[0] (rising edge-triggered flip-flop clocked by clk)
Endpoint: sum_reg[21]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Point	Incr	Path
clock clk (rise edge)	2.85	2.85
clock network delay (ideal)	0.00	2.85
b_q_reg[0]/CLK (dffs2)	0.00	2.85 r
sum_reg[21]/DIN (dffles1)	0.00	8.07 r
data arrival time		8.07
clock clk (rise edge)	8.55	8.55
clock network delay (ideal)	0.00	8.55
sum_reg[21]/CLK (dffles1)	0.00	8.55 r
library setup time	-0.48	8.07
data required time		8.07
data required time		8.07
data arrival time		-8.07
slack (MET)		0.00

$$8.55 - 2.85 = 5.7$$

```
create_clock -name "clk" -period 5.7 -waveform { 0 2.85 } { clk }
```

Startpoint: b_q_reg[0] (rising edge-triggered flip-flop clocked by clk)
Endpoint: sum_reg[21]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
b_q_reg[0]/CLK (dffs2)	0.00	0.00 r
sum_reg[21]/DIN (dffles1)	0.00	5.22 r
data arrival time		5.22
clock clk (rise edge)	5.70	5.70
clock network delay (ideal)	0.00	5.70
sum_reg[21]/CLK (dffles1)	0.00	5.70 r
library setup time	-0.48	5.22
data required time		5.22
data required time		5.22
data arrival time		-5.22
slack (MET)		0.00

$$5.70 - 0.00 = 5.7$$

Min. CLK Period = 5.7ns

Max. CLK Frequency = 175.44MHz

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 - Do the pipelining with the Gate level description & Check the functionality
 - Measure the critical path delay & circuit area & power
3. 2-stage pipelined 22-bit Square Root Carry Select Adder
 - Do the pipelining with the Gate level description & Check the functionality
 - Measure the critical path delay & circuit area & power
4. Synthesis Results Analysis
 - Check the experimental results (max. clock cycle, area, power)

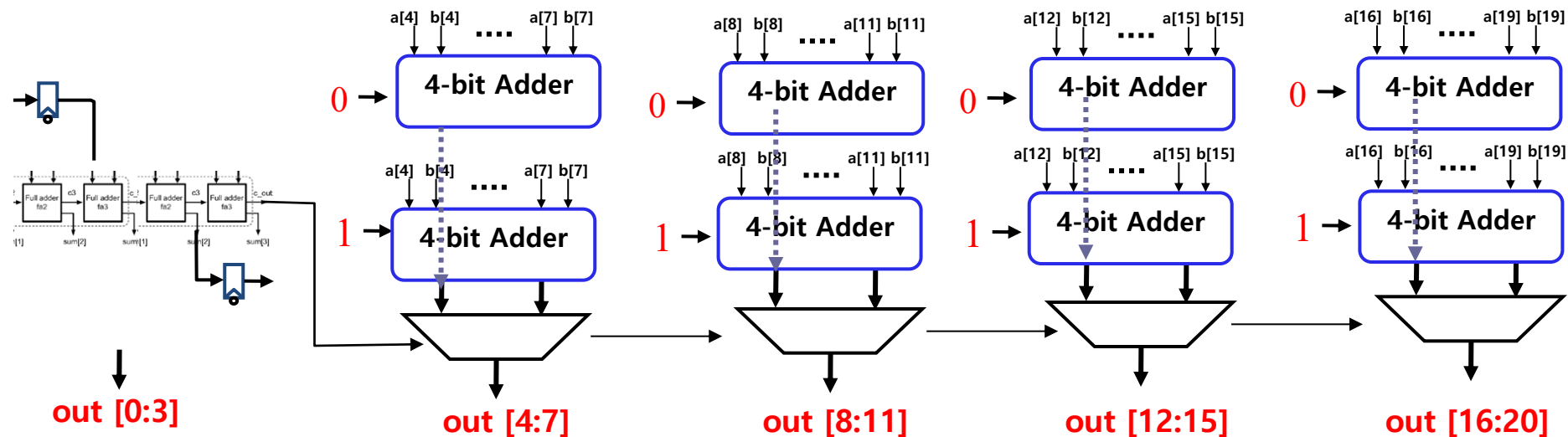
2-stage pipelined Carry Select Adder

Design 2-stage pipelined **20-bit Carry Select Adder**

What is the **Path Delay** for each path? (FA = MUX)

Reduce the clock until it is violated !

What is **MAXIMUM Clock Frequency**?



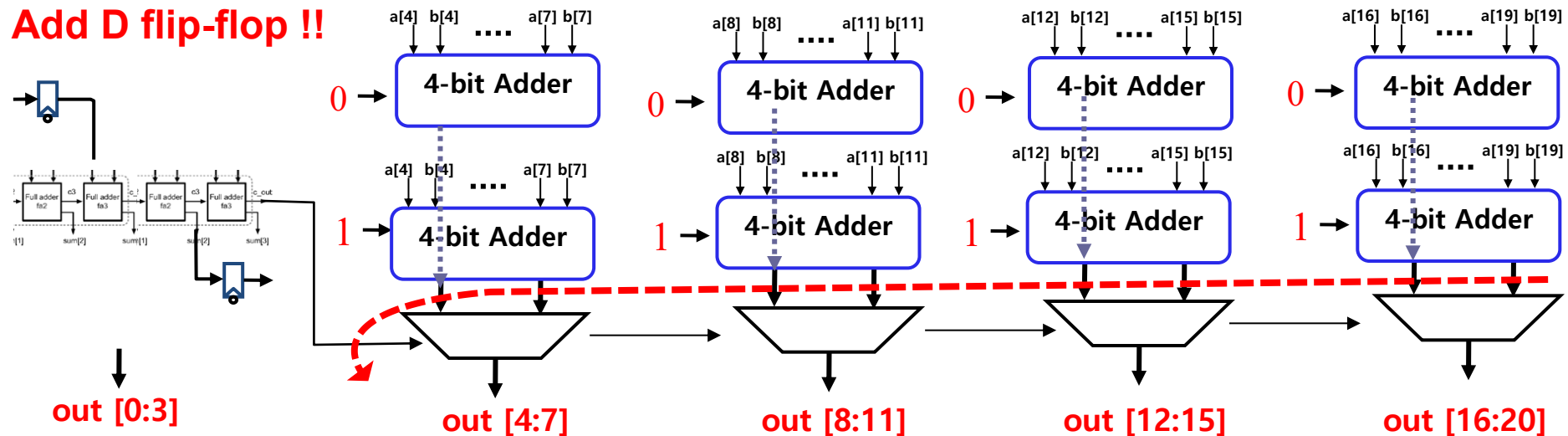
2-stage pipelined Carry Select Adder

Design 2-stage pipelined **20-bit Carry Select Adder**

What is the **Path Delay** for each path? (FA = MUX)

Reduce the clock until it is violated !

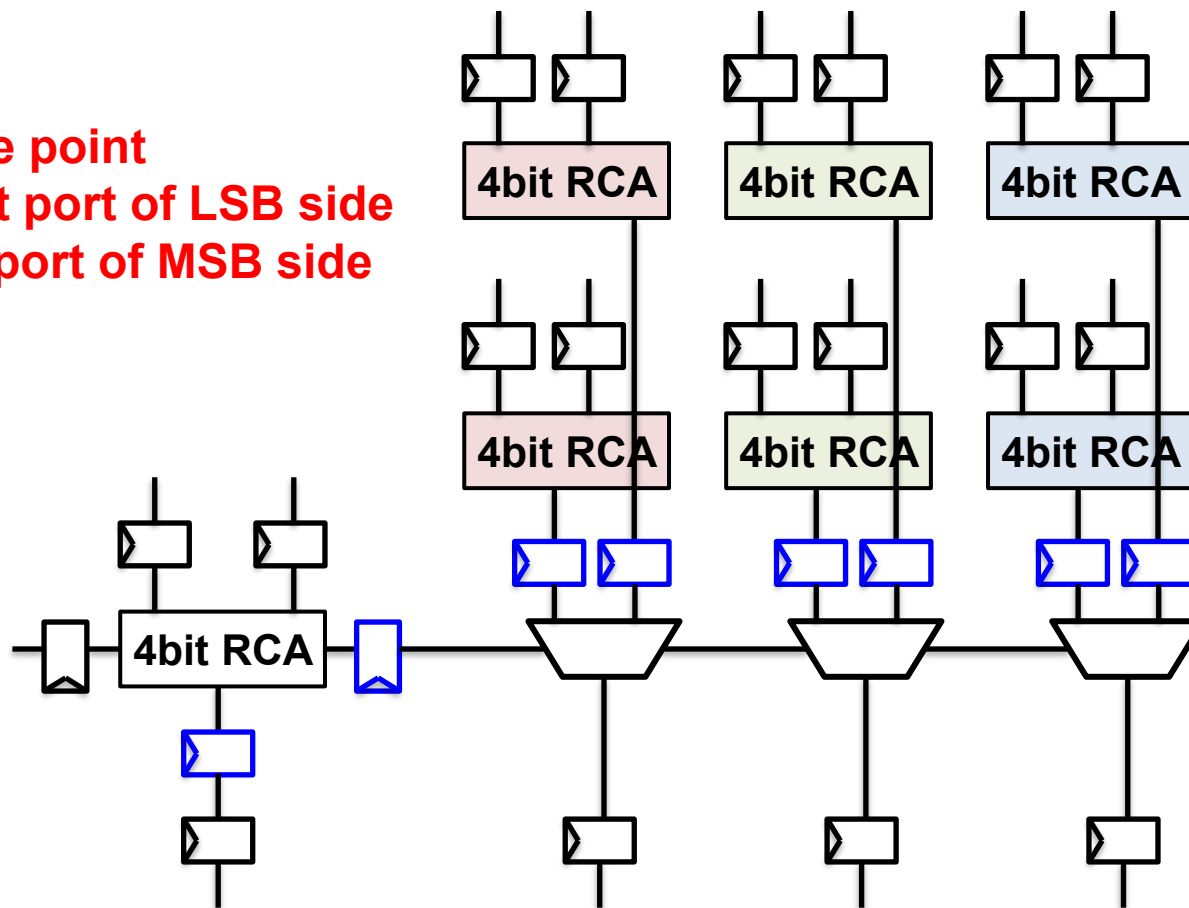
What is **MAXIMUM Clock Frequency**?



Example

Example of 2-stage pipelined 16-bit Carry Select Adder

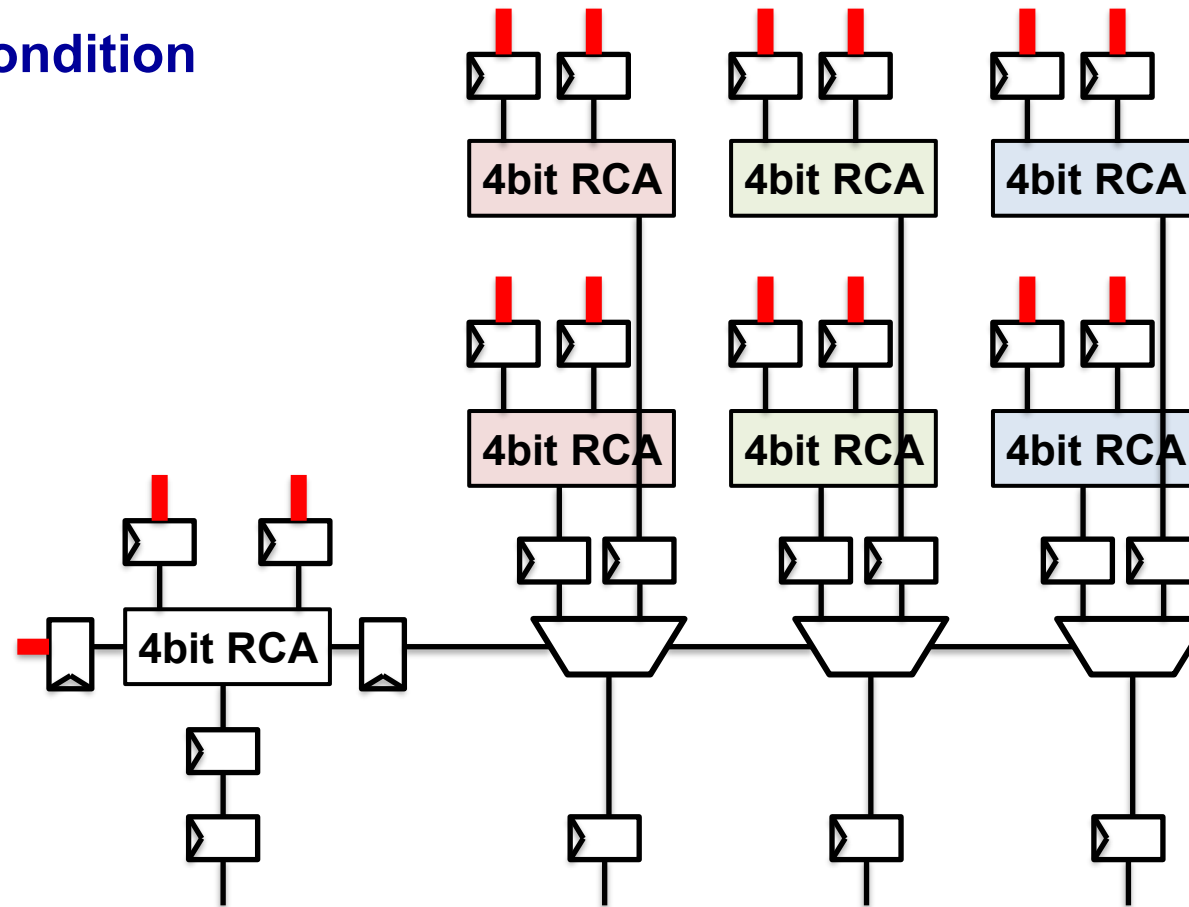
1. Add D flip-flop in middle point
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Example

Example of 2-stage pipelined 16-bit Carry Select Adder

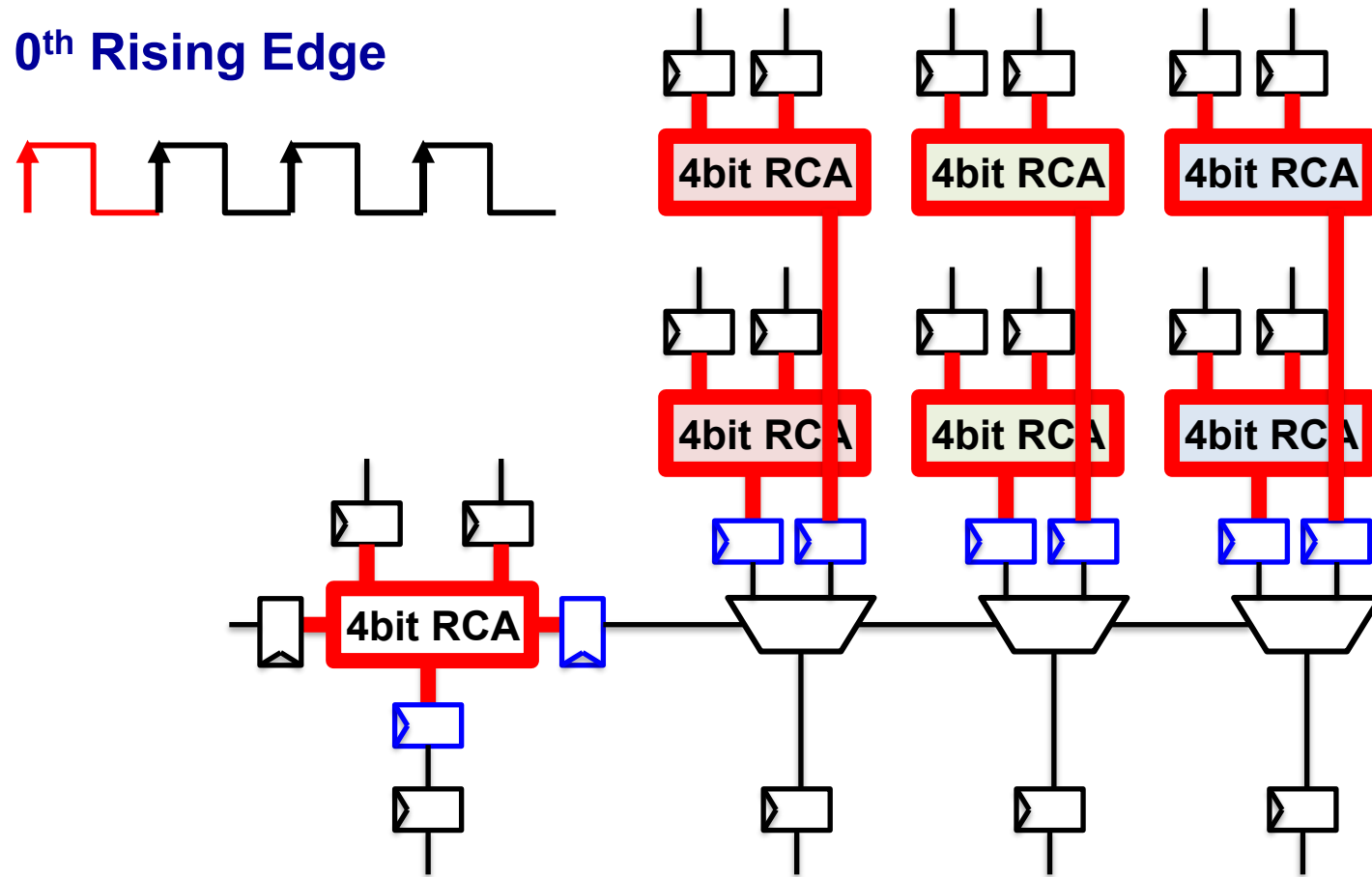
Initial Condition



Example

Example of 2-stage pipelined 16-bit Carry Select Adder

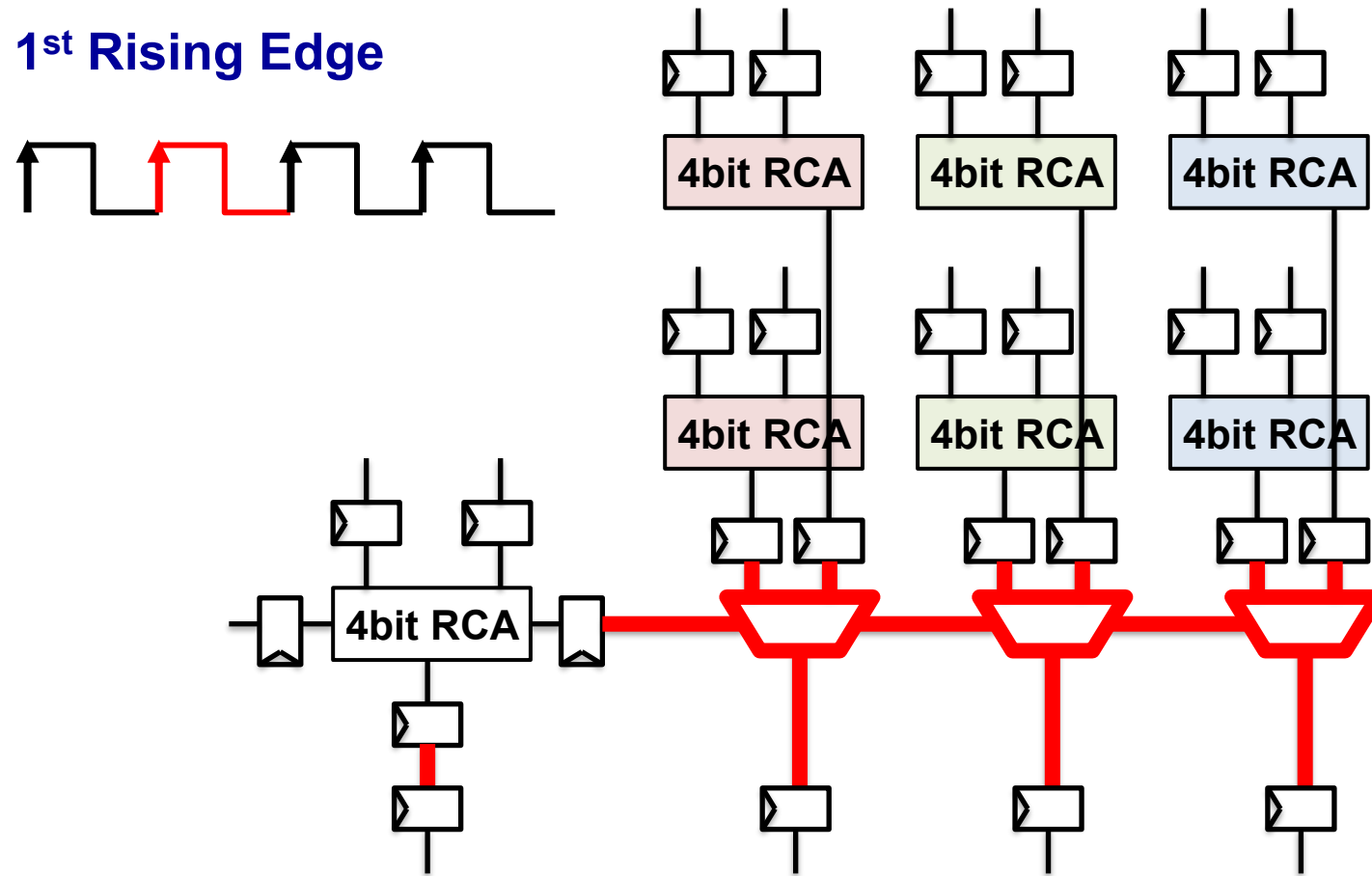
0th Rising Edge



Example

Example of 2-stage pipelined 16-bit Carry Select Adder

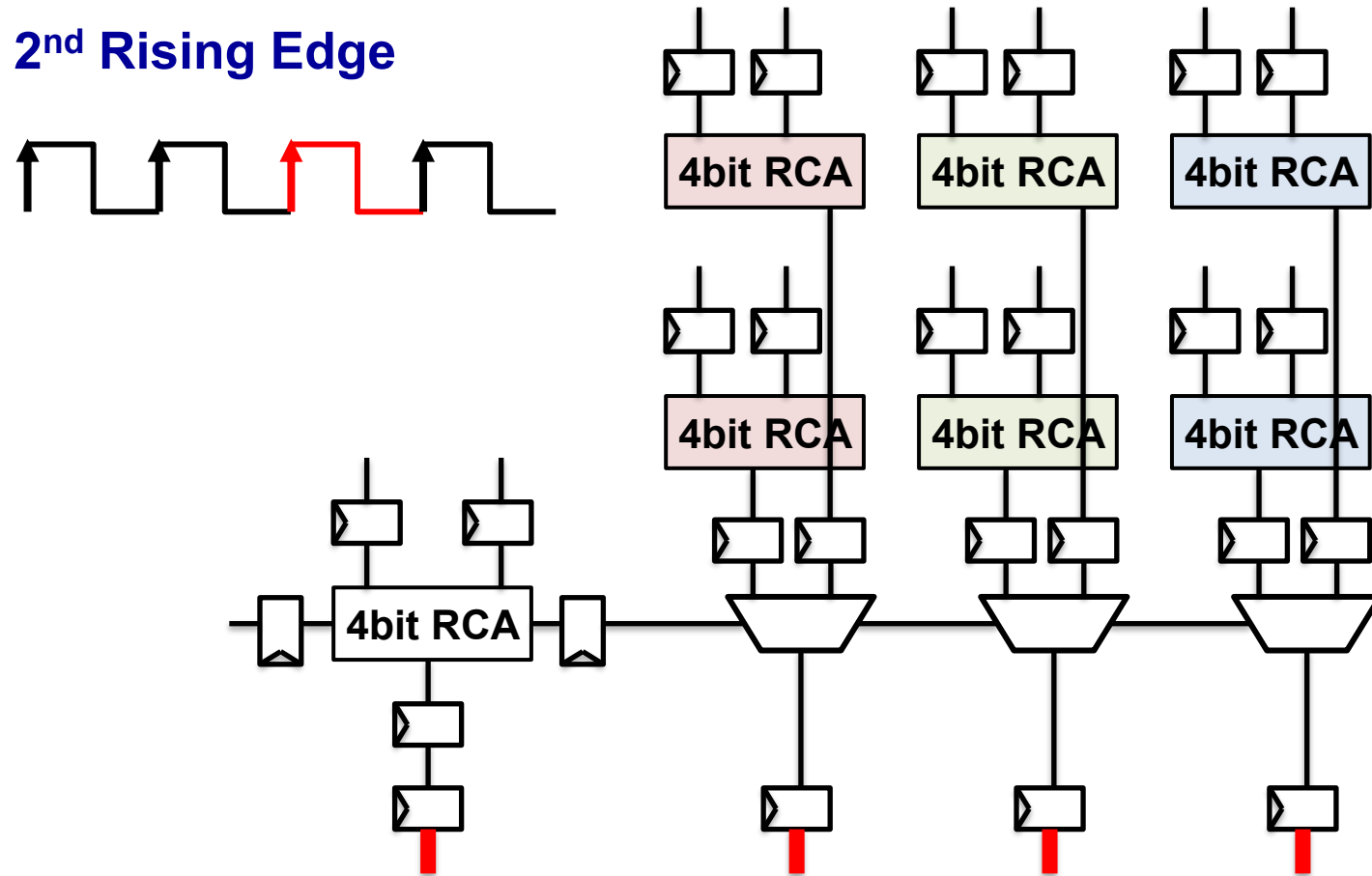
1st Rising Edge



Example

Example of 2-stage pipelined 16-bit Carry Select Adder

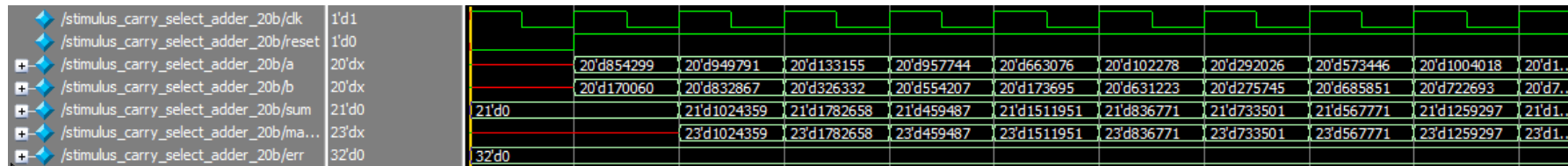
2nd Rising Edge



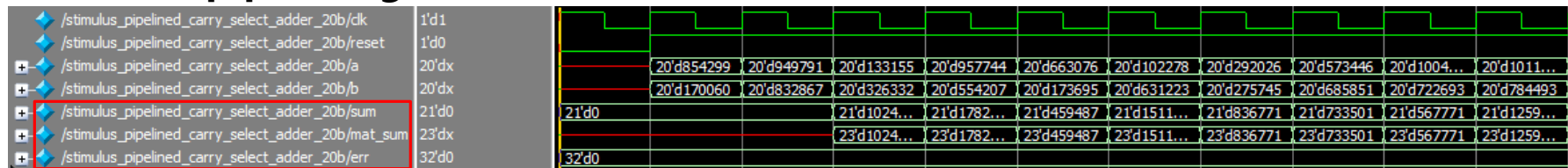
Function Matching

• Simulation results 20-bit Carry Select Adder

- without pipelining : Don't need to check this waveform



- with pipelining : Please check this waveform



Check outputs of two adders and Matlab results are 'same'

Check the value of error counting number equals 'zero'

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4. **Synthesis Results Analysis**
 - Check the experimental results (max. clock cycle, area, power)

2-stage pipelined Square Root Carry Select Adder

Design 2-stage pipelined 22-bit Square Root Carry Select Adder

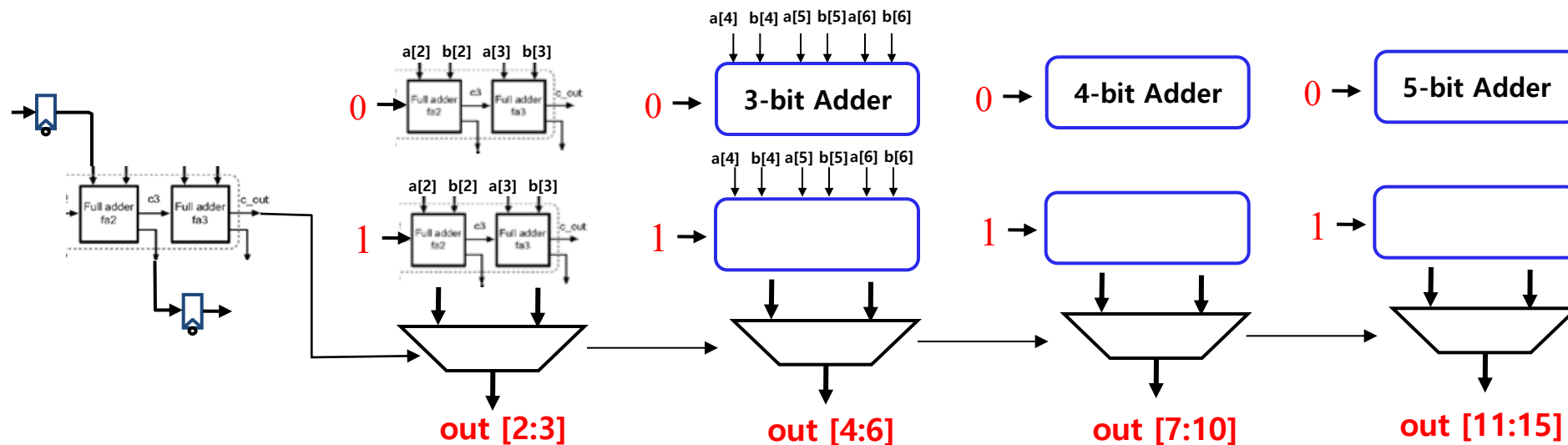
Reduce the clock until it is violated !

What is **MAXIMUM** Clock cycle ?

What is **AREA** Change ?

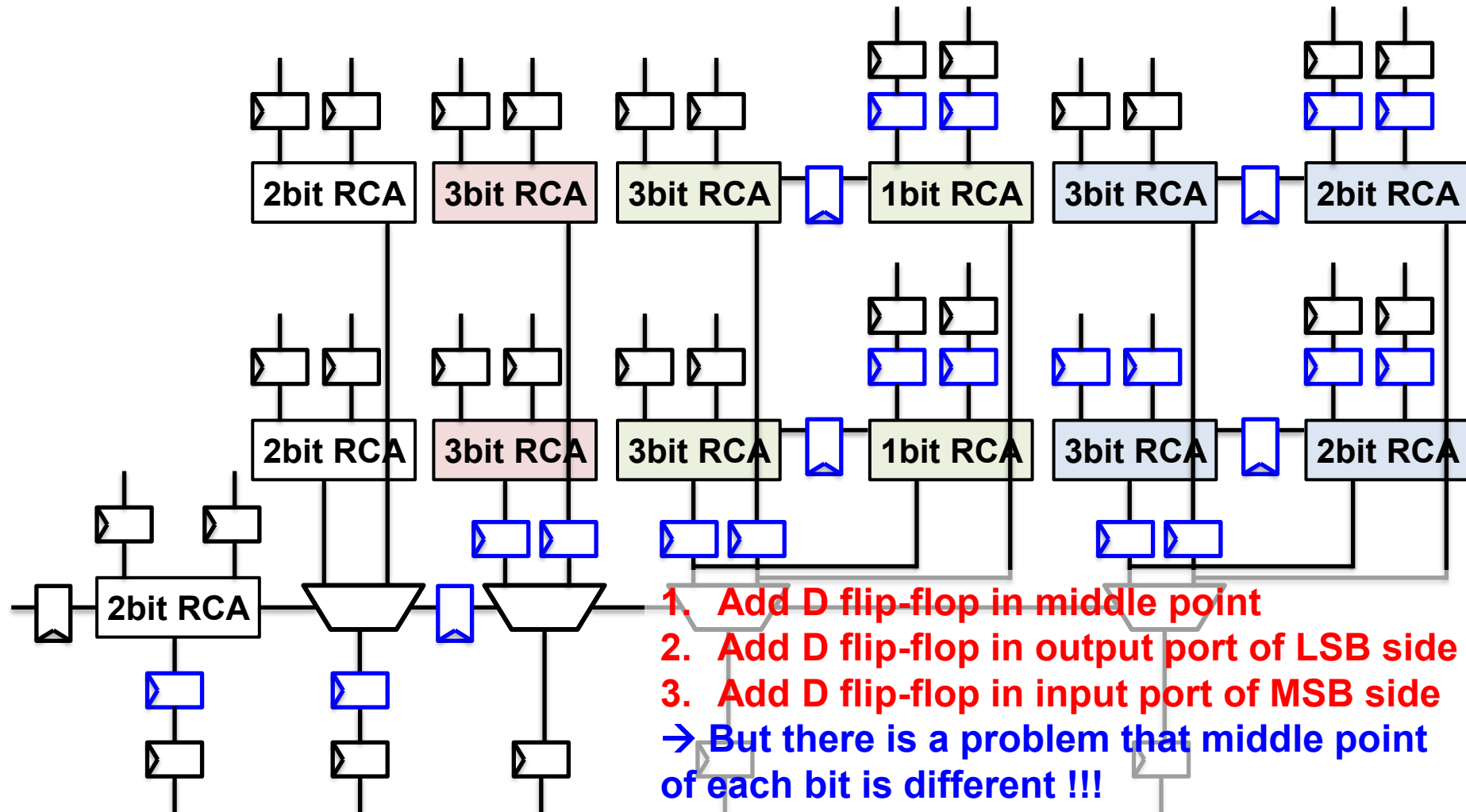
Critical Path ?
Other Paths ?

ex) 16-bit Square Root Carry Select Adder



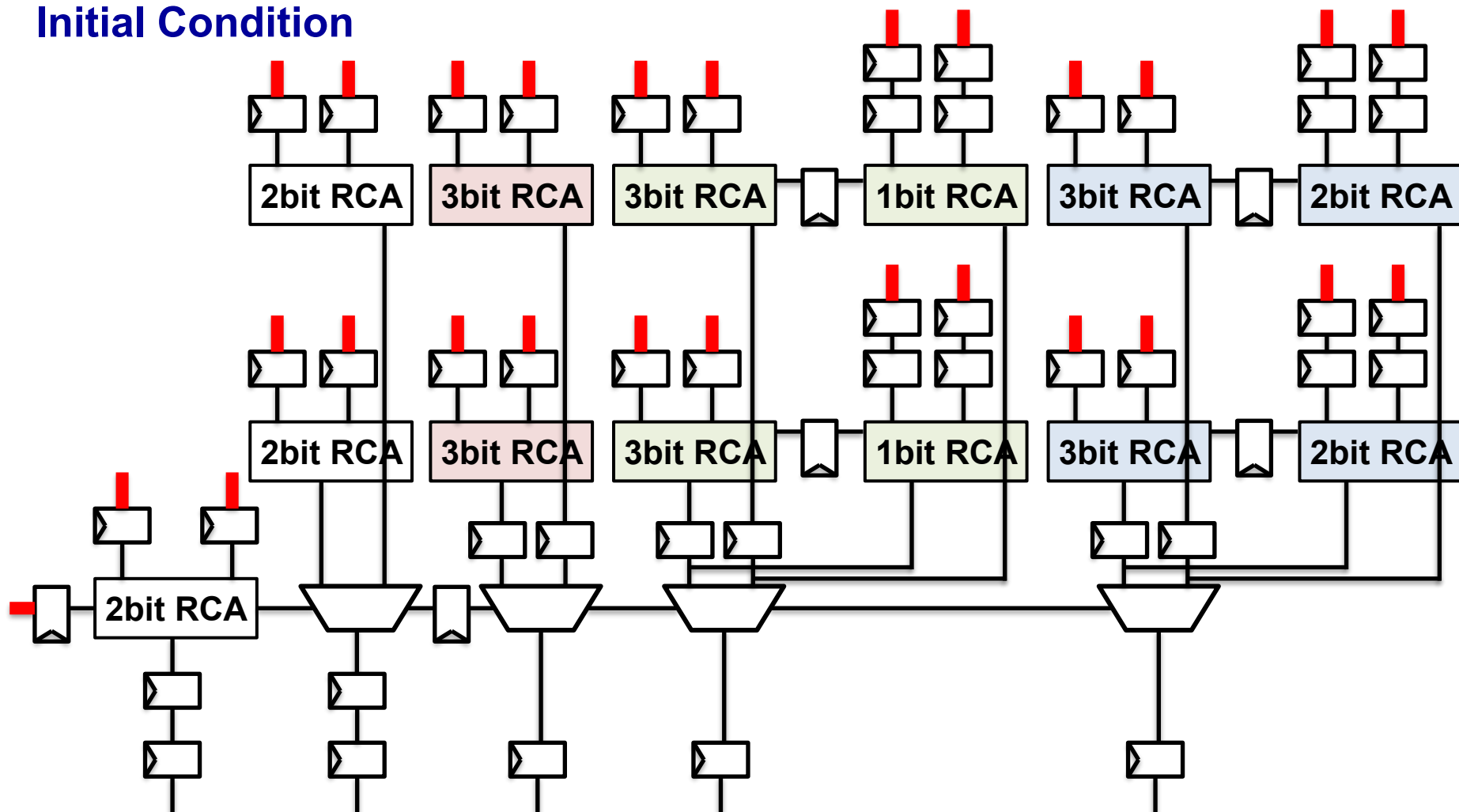
Critical Path = 2 FA + 4 MUX = 6 Unit delay

Example



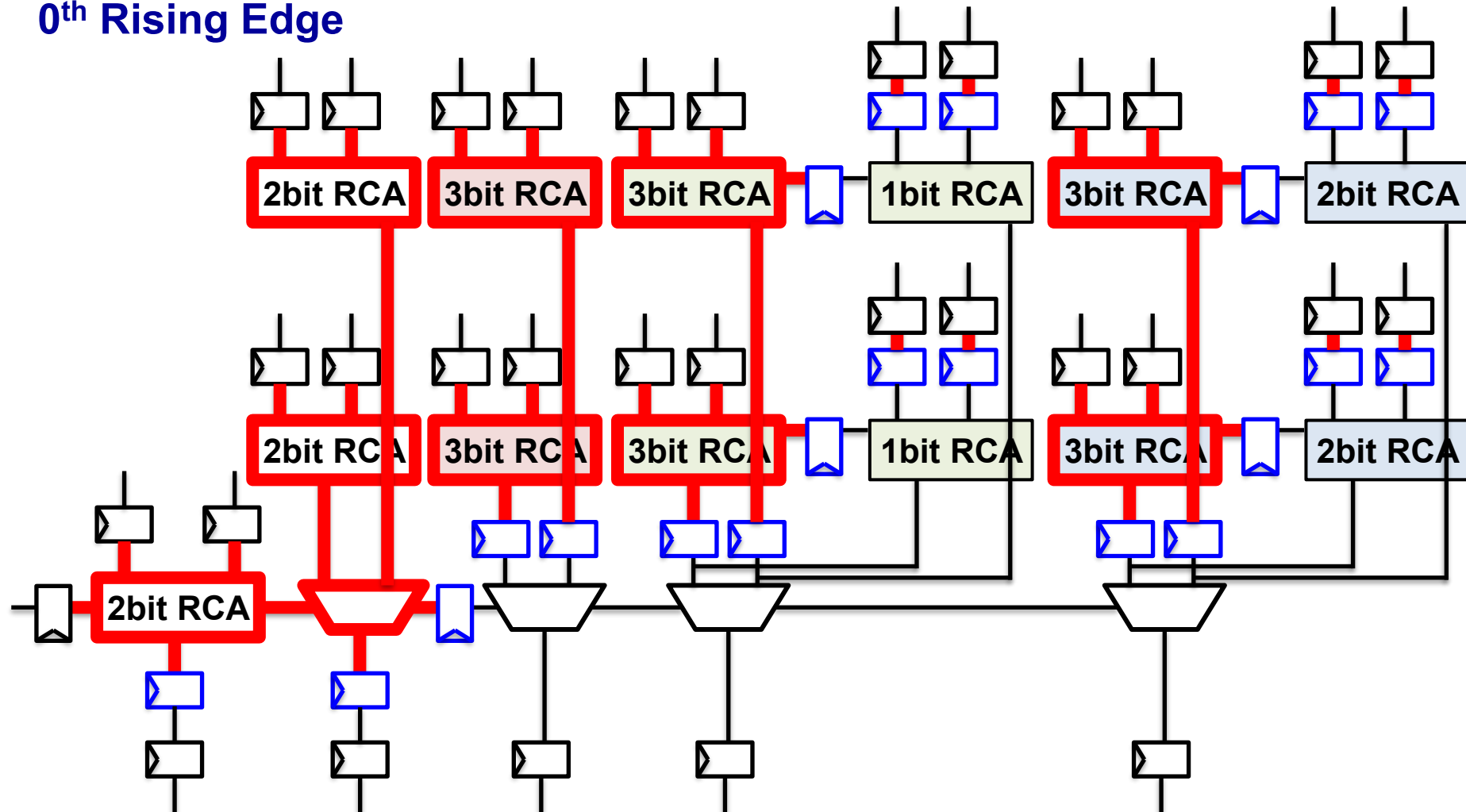
Example

Initial Condition



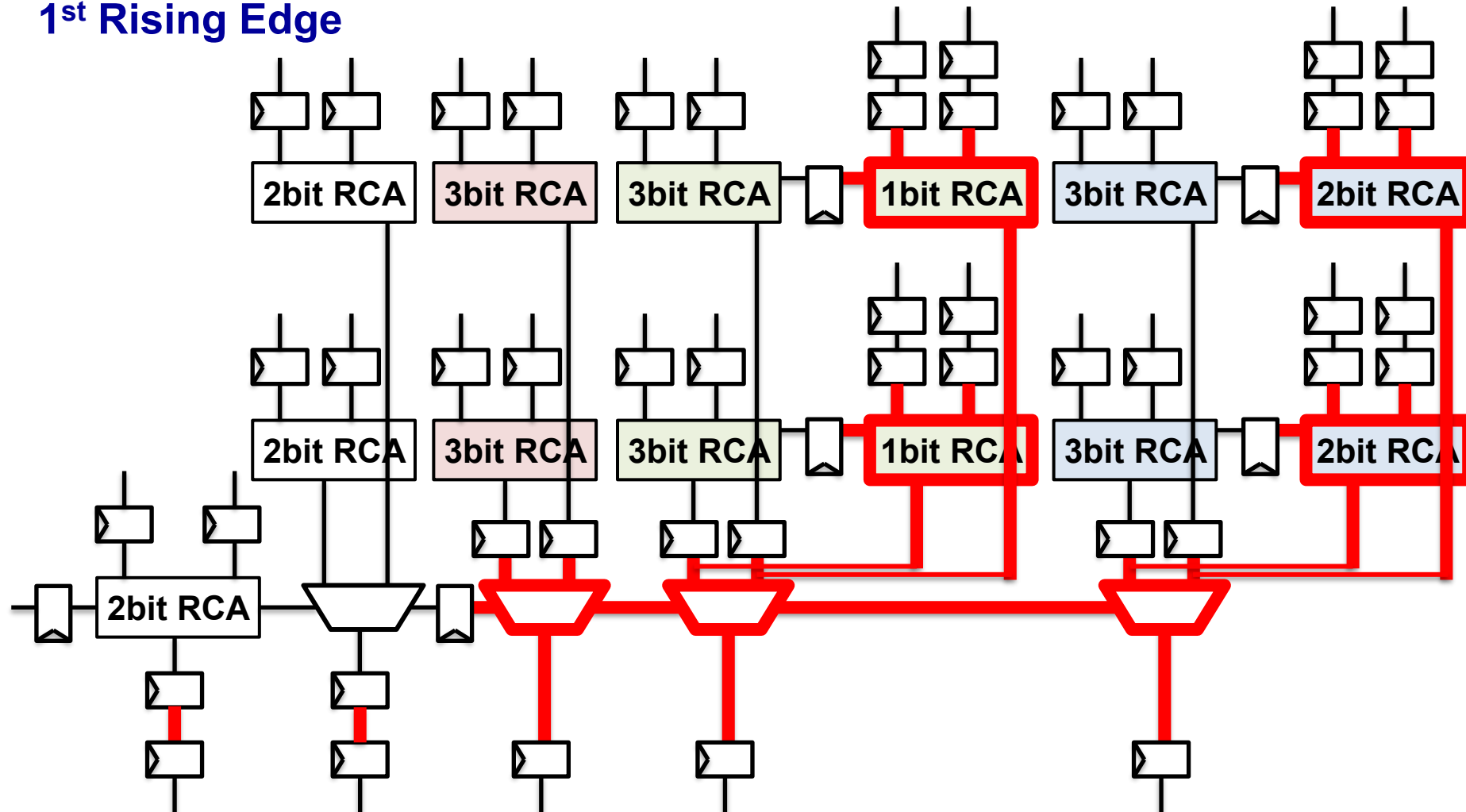
Example

0th Rising Edge



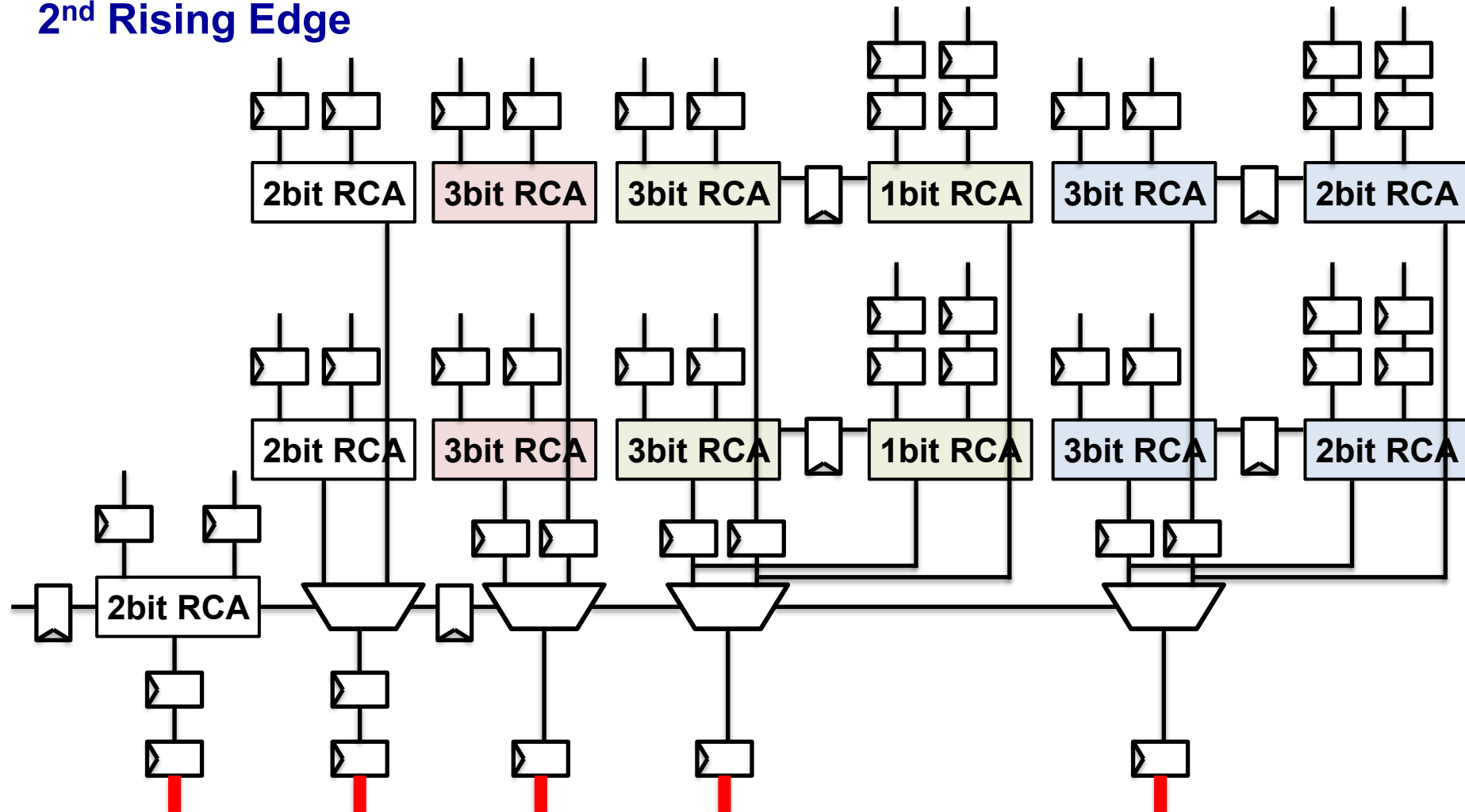
Example

1st Rising Edge



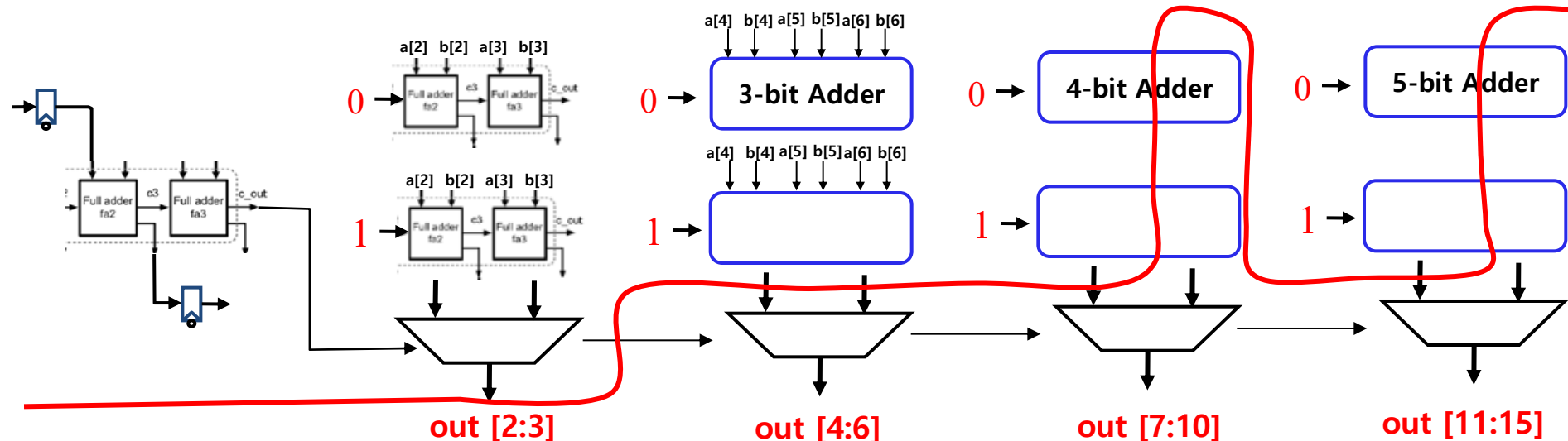
Example

2nd Rising Edge



2-stage pipelined Square Root Carry Select Adder

Design 2-stage pipelined 22-bit Square Root Carry Select Adder
ex) 16-bit Square Root Carry Select Adder



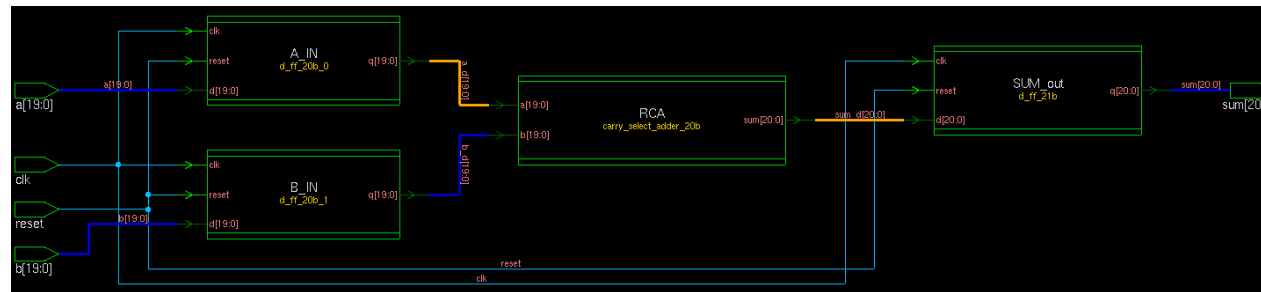
Do the 2-stage pipelining on the
22-bit Square Root Carry Select Adder
on your own !!

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 - Measure the critical path delay & circuit area & power
4. **Synthesis Results Analysis**
 - Check the experimental results (max. clock cycle, area, power)

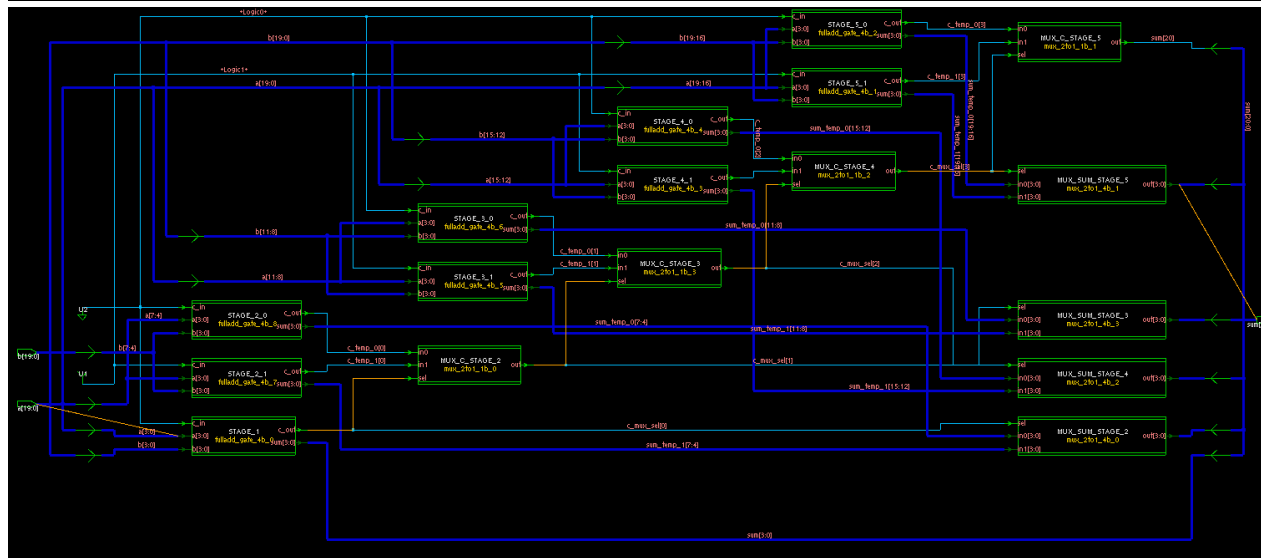
Find the maximum clock cycle

- Synthesis results** 22-bit Carry Select Adder



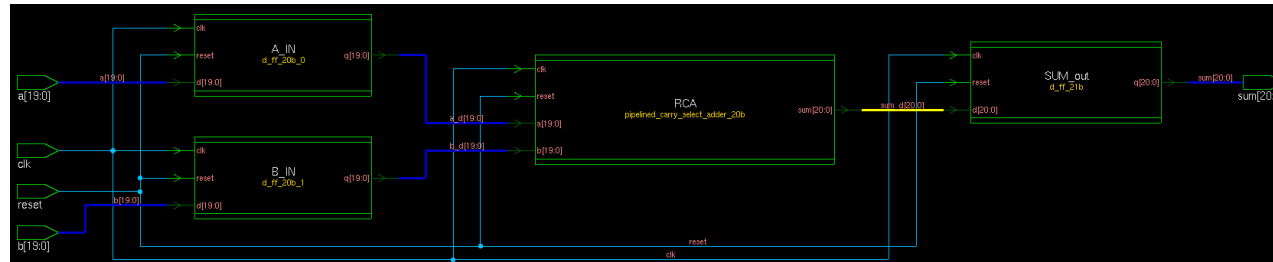
without pipelining

Don't need to
check this schematic



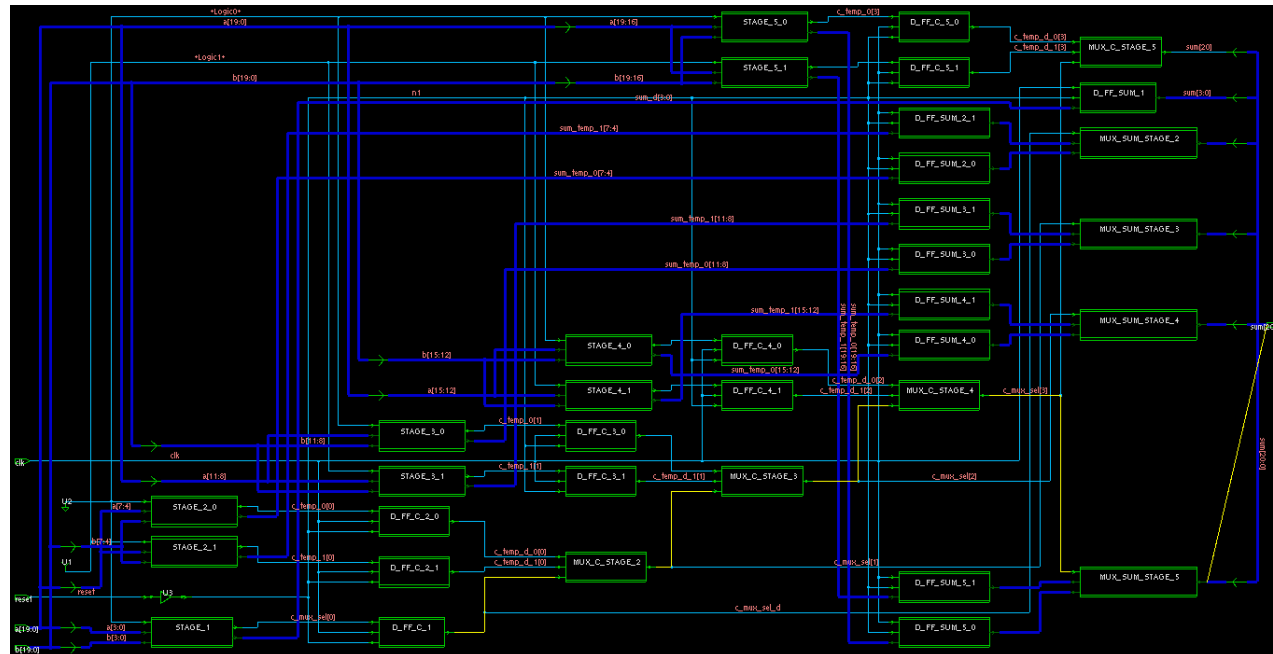
Find the maximum clock cycle

- Synthesis results** 22-bit Carry Select Adder



with pipelining

Please check
this schematic



Ctrl+H
to highlight
critical path

Check the synthesis results

• Critical path delay, Circuit area & Power

report_timing

```
*****
Report : timing
-path full
-delay max
-max_paths 1
Design : top_pipelined_carry_select_adder_20b
Version: Z-2007.03-SP4
Date : Fri Apr 1 19:09:20 2016
*****

Point                                Incr      Path
-----
clock clk (rise edge)                0.00      0.00
clock network delay (ideal)           0.00      0.00
RCA/D_FF_C_1/q_reg/CK (fd1qd1_hd)     0.00      0.00 r
RCA/D_FF_C_1/q_reg/0 (fd1qd1_hd)      0.61      0.61 r
RCA/D_FF_C_1/q (d_ff_1b_0)            0.00      0.61 r
RCA/MUX_C_STAGE_2/0 (mux_2to1_1b_0)   0.00      0.61 r
RCA/MUX_C_STAGE_2/U1/Y (ivd1_hd)       0.17      0.77 f
RCA/MUX_C_STAGE_2/U2/Y (scg2d2_hd)     0.42      1.20 f
RCA/MUX_C_STAGE_2/out (mux_2to1_1b_0)  0.00      1.20 f
RCA/MUX_C_STAGE_3/0 (mux_2to1_1b_3)   0.00      1.20 f
RCA/MUX_C_STAGE_3/U1/Y (ivd1_hd)       0.13      1.33 r
RCA/MUX_C_STAGE_3/U2/Y (scg2d2_hd)     0.34      1.67 r
RCA/MUX_C_STAGE_3/out (mux_2to1_1b_3)  0.00      1.67 r
RCA/MUX_C_STAGE_4/0 (mux_2to1_1b_2)   0.00      1.67 r
RCA/MUX_C_STAGE_4/U1/Y (ivd1_hd)       0.13      1.79 f
RCA/MUX_C_STAGE_4/U2/Y (scg2d2_hd)     0.41      2.20 f
RCA/MUX_C_STAGE_4/out (mux_2to1_1b_2)  0.00      2.20 f
RCA/MUX_SUM_STAGE_5/0 (mux_2to1_4b_1)  0.00      2.20 f
RCA/MUX_SUM_STAGE_5/U1/Y (ivd1_hd)     0.27      2.47 r
RCA/MUX_SUM_STAGE_5/U2/Y (scg2d2_hd)   0.26      2.74 r
RCA/MUX_SUM_STAGE_5/out[0] (mux_2to1_4b_1) 0.00      2.74 r
RCA/sum[16] (pipelined_carry_select_adder_20b) 0.00      2.74 r
SUM_out/d[16] (d_ff_21b)              0.00      2.74 r
SUM_out/U3/Y (add2d2_hd)              0.15      2.89 r
SUM_out/q_reg[16]/D (fd1qd1_hd)       0.00      2.89 r
data arrival time                      2.89

clock clk (rise edge)                3.00      3.00
clock network delay (ideal)           0.00      3.00
SUM_out/q_reg[16]/CK (fd1qd1_hd)      0.00      3.00 r
library setup time                    -0.11      2.89
data required time                     2.89
data required time                     2.89
data arrival time                      -2.89
slack (MET)                           0.00
```

report_area

```
*****
Report : area
Design : top_pipelined_carry_select_adder_20b
Version: Z-2007.03-SP4
Date : Fri Apr 1 19:11:27 2016
*****

Number of ports: 63
Number of nets: 124
Number of cells: 4
Number of references: 4

Combinational area: 514.667088
Noncombinational area: 565.332996
Net Interconnect area: 6.245844

Total cell area: 1080.000122
Total area: 1086.245966
```

report_power

```
*****
Report : power
-analysis_effort low
Design : top_pipelined_carry_select_adder_20b
Version: Z-2007.03-SP4
Date : Fri Apr 1 19:13:05 2016
*****

Global Operating Voltage = 1.05
Power-specific unit information :
Voltage Units = 1V
Capacitance Units = 1.000000pf
Time Units = 1ns
Dynamic Power Units = 1mW (derived from V,C,T units)
Leakage Power Units = 1mW

Cell Internal Power = 886.9149 uW (80%)
Net Switching Power = 215.2427 uW (20%)
-----
Total Dynamic Power = 1.1022 mW (100%)
Cell Leakage Power = 2.1086 uW
```

Using Random Vector!

Check results and Submit report

- **Check the synthesis result**
 - **Show waveforms & synthesis results (schematic)**
 - **Show timing, area, power report & zip(.v file)**
 - **Check the highlighted critical path**
- **Submit result report**
 - **Show the timing diagram and the above synthesis results**
 - **Results analysis**
- **Due : Before the next class to your group TA**
(담당 조교에게 제출 바랍니다)